Portfolio Optimisation

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Why Risk Matters

	Return	*Risk*
Fund A	30%	High
Fund B	15%	Low

★ The goal is not just to maximise return, but to find the best balance between risk and return for each investor individually.

Utility

What does receiving £100 mean to you?

- ★ To a student it could change their week.
- ★ To a billionaire it likely wouldn’t affect their day.

Utility is a measure of the satisfaction an investor derives from an outcome, it is unique to each individual.

Sharpe Ratio

Several metrics exist to measure risk-adjusted portfolio performance, each capturing a different aspect.

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_\Pi}$$

R_p *Expected return of portfolio*

R_f *Risk-free return*

σ_Π *Standard deviation of portfolio*

- ★ How much excess return you're earning per unit of risk taken.
- ★ Maximising the Sharpe Ratio connects directly to the efficient frontier and capital market line.

Alternative metrics to quantify the performance of a portfolio

What might people value instead of pure profit

This informs on their utility function

Shapes of utility functions and

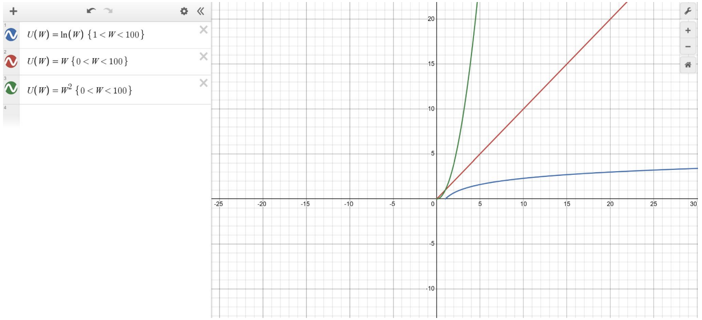
A utility function describes the relationship between wealth and utility.

A simple form is:

where:

W = wealth

U(W) = utility from wealth



Investor Type	Shape	Explanation
Risk- Averse	Concave	Utility increases at a decreasing rate
Risk-Neutral	Linear	Utility increases at a constant rate
Risk-Seeking	Convex	Utility increases at an increasing rate

What matters is not the exact formula, but the shape of the utility function. The shape reflects how investors value additional wealth and therefore reflects their attitude toward risk.

Representing a portfolio as a vector

For any person, we say that they have $Wealth = 1$

And in a market of n stocks we represent a portfolio by the vector

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_n \end{pmatrix}^T$$

Where λ_i represents a fraction of total wealth allocated to stock i

And in this model we have $\sum_{i=1}^{n} \lambda_i = Wealth$

How can we measure risk?

Looking at a stock independently, we can use variance to understand the risk of holding it.

However we are in a stock market. We might wonder, do particular stocks within the market have bearings on one another? If so, by how much?

Covariance in a stock market

Consider stocks Apple (AAPL) and Microsoft (MSFT). Is there any relation between the two? What happens to one if the other rises? What happens to one if the other falls? Both companies,

- Attract many of the same investors
- Are major holdings in index funds like the S&P 500 and Nasdaq-100
- Are major players in AI

Clearly there is a correlation. Such can exist between other stocks too.

Statistically we measure this using **covariance**.

The Covariance Matrix

We use the introduce **covariance matrix**,

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

Where each entry σ_{ij} denotes the covariance between stocks i and j

Risk of a portfolio

With the covariance matrix, $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$

and portfolio allocation, $\boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_n \end{pmatrix}^T$

we define the risk of a portfolio to be: $\sigma_{\Pi}^2 = \Lambda^T \Sigma \, \Lambda$

We want to answer:

- Can we find a way to minimise our portfolio risk?
- How should we decide the balance between a low risk and a high Sharpe ratio?

Optimisation method: Lagrange Multipliers

The method of lagrange multipliers is a strategy for finding the local maxima or minima of a function subject to equation constraints.

The idea is to modify a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

The general method is:

1. Identify what you are trying to optimise, e.g. the function $f(\vec{x})$
2. Identify your constraints e.g. $h_1(\vec{x}) = 1$, $h_2(\vec{x}) = 5$ Note: can be inequalities – KKT conditions
3. Express your constraints in the form $g(\vec{x}) = 0$ e.g. $g_1(\vec{x}) := h_1(\vec{x}) - 1 = 0$ and $g_2(\vec{x}) := h_2(\vec{x}) - 5 = 0$
4. Find Lagrangian Equation, $\mathcal{L}(\vec{x}, \lambda_1, \lambda_2) = f(\vec{x}) - \lambda_1 g_1(\vec{x}) - \lambda_2 g_2(\vec{x})$
5. Take the gradient, set to 0: $\nabla_{\vec{x}} \mathcal{L}(\vec{x}, \lambda_1, \lambda_2) = \nabla f(\vec{x}) - \lambda_1 \nabla g_1(\vec{x}) - \lambda_2 \nabla g_2(\vec{x}) = 0$
6. Solve $\nabla f(\vec{x}) - \lambda_1 \nabla g_1(\vec{x}) - \lambda_2 \nabla g_2(\vec{x}) = 0$ along with $g_1(\vec{x}) = 0$ and $g_2(\vec{x}) = 0$

Example of using Lagrange Multipliers

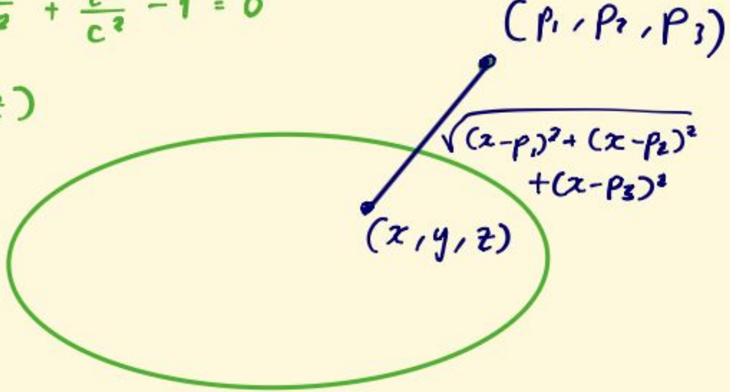
For example, if we want to find the point lying on an ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

closest to an external point (p_1, p_2, p_3), we want to minimise distance, i.e.

we want to minimise $f(x, y, z) = (x - p_1)^2 + (y - p_2)^2 + (z - p_3)^2$

subject to the following constraint $g(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$

$$\mathcal{L}(x, y, z, \lambda) = f(x, y, z) - \lambda g(x, y, z)$$



we solve alongside

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \end{cases}$$

Minimising risk using Lagrange multipliers: Setup

Consider an n-stock portfolio, containing stocks $S_1, \ldots S_n$

Let us define $\vec{\lambda} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$ as a vector of portfolio weights

$\vec{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}$ as a vector of expected returns of each stock

$\Sigma = \begin{pmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \\ \sigma_{n1} & & \sigma_{nn} \end{pmatrix}$ as the covariance matrix

And we will write $\vec{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$

Our aim is to minimise portfolio risk, $\sigma_{\Pi}^2 = \vec{\Lambda}^{\mathsf{T}} \Sigma \vec{\Lambda}$
subject to the following constraints:

[1] $\vec{\mu}^{\,\prime}\vec{\Lambda} = r$ (the expected returns of the portfolio, r, must equals the weighted returns of each stock)

[2] $\vec{1}^{\,\prime}\vec{\Lambda} = 1$ (the weight of all the stocks must add up to 1)

We might also include the constraints

[3] $\lambda_i \geq 0$ (we cannot short a stock)

but we can deal with this later if needed

Modelling Assumptions

- Convexity of variance– this ensures that the variance is minimised

- Invertibility of covariance matrix– this is almost certainly the case, it is not the case only when the returns of a stock is always identically equal to the returns to another stock in portfolio

Formulating our Lagrangian Equation

Then, we can convert these constraints into [1] $\vec{\mu}^T\vec{\Lambda} - r = 0$ and [2] $\vec{1}^T\vec{\Lambda} - 1 = 0$

Applying the Lagrangian:

$$\mathcal{L}(\vec{\Lambda}, \alpha, \beta) = \sigma_\Pi^2 - \alpha(\vec{\mu}^T\vec{\Lambda} - r) - \beta(\vec{1}^T\vec{\Lambda} - 1)$$

$$= \vec{\Lambda}^T \Sigma \vec{\Lambda} - \alpha(\vec{\mu}^T\vec{\Lambda} - r) - \beta(\vec{1}^T\vec{\Lambda} - 1)$$

Then we differentiate wrt $\vec{\Lambda}$ (aka take the gradient)

$$\nabla_\Lambda \mathcal{L}(\vec{\Lambda}, \alpha, \beta) = 2\Sigma\vec{\Lambda} - \alpha\vec{\mu} - \beta\vec{1}$$

Setting this to zero:

$$2\Sigma\vec{\Lambda} - \alpha\vec{\mu} - \beta\vec{1} = 0$$

$$\Rightarrow \quad 2\Sigma\vec{\Lambda} = \alpha\vec{\mu} + \beta\vec{1}$$

Assuming Σ is invertible (Note: This is a non-trivial assumption, will be explained after.)

$$\boxed{\vec{\Lambda} = \frac{1}{2}\Sigma^{-1}(\alpha\vec{\mu} + \beta\vec{1})}$$

Finding α and β

Now we want to eliminate our lagrange multipliers α and β
We can substitute our $\vec{\Lambda}$ solution into our first constraint

[1] $\vec{\mu}^{\top}\vec{\Lambda} = r$

$$\Rightarrow \quad \vec{\mu}^{\top}\left[\frac{1}{2}\Sigma^{-1}(\alpha\vec{\mu} + \beta\vec{1})\right] = r$$

$$\Rightarrow \quad \alpha\vec{\mu}^{\top}\Sigma^{-1}\vec{\mu} + \beta\vec{\mu}^{\top}\Sigma^{-1}\vec{1} = 2r$$

Then we substitute our $\vec{\Lambda}$ solution into our second constraint

[2] $\vec{1}^{\top}\vec{\Lambda} = 1$

$$\Rightarrow \quad \vec{1}^{\top}\left[\frac{1}{2}\Sigma^{-1}(\alpha\vec{\mu} + \beta\vec{1})\right] = 1$$

$$\Rightarrow \quad \alpha\vec{1}^{\top}\Sigma^{-1}\vec{\mu} + \beta\vec{1}^{\top}\Sigma^{-1}\vec{1} = 2$$

Finding the portfolio weight vector

Now we have simultaneous equations

$$\begin{cases} \alpha \vec{\mu}^T \Sigma^{-1} \vec{\mu} + \beta \vec{\mu}^T \Sigma^{-1} \vec{1} = 2r \\ \alpha \vec{1}^T \Sigma^{-1} \vec{\mu} + \beta \vec{1}^T \Sigma^{-1} \vec{1} = 2 \end{cases}$$

Solving them is excruciatingly painful, so I will just show our results:

$$\boxed{\alpha = \frac{2[r \vec{1}^T \Sigma^{-1} \vec{1} - \vec{1}^T \Sigma^{-1} \vec{\mu}]}{(\vec{1}^T \Sigma^{-1} \vec{1})(\vec{\mu}^T \Sigma^{-1} \vec{\mu}) - (\vec{1}^T \Sigma^{-1} \vec{\mu})^2}}$$

$$\boxed{\beta = \frac{2[\vec{\mu}^T \Sigma^{-1} \vec{\mu} - r \vec{1}^T \Sigma^{-1} \vec{\mu}]}{(\vec{1}^T \Sigma^{-1} \vec{1})(\vec{\mu}^T \Sigma^{-1} \vec{\mu}) - (\vec{1}^T \Sigma^{-1} \vec{\mu})^2}}$$

Plugging in the α and β into the $\vec{\Lambda}$ vector gives something big and ugly:

$$\vec{\Lambda} = \frac{1}{2} \Sigma^{-1} (\alpha \vec{\mu} + \beta \vec{1})$$

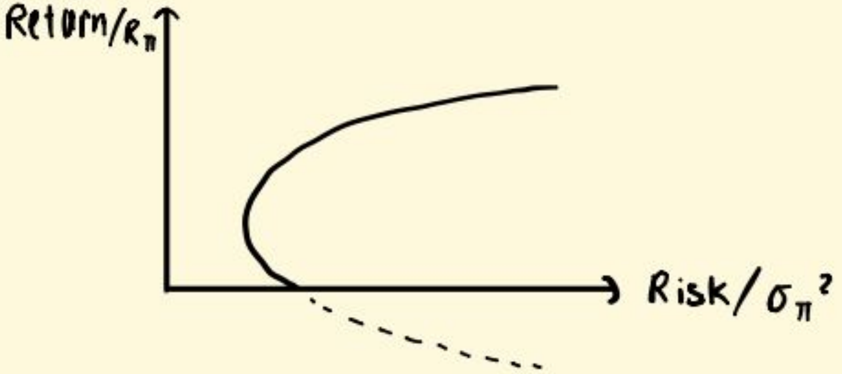
$$\Rightarrow \boxed{\vec{\Lambda} = \Sigma^{-1} \left[\left[\frac{[r \vec{1}^T \Sigma^{-1} \vec{1} - \vec{1}^T \Sigma^{-1} \vec{\mu}]}{(\vec{1}^T \Sigma^{-1} \vec{1})(\vec{\mu}^T \Sigma^{-1} \vec{\mu}) - (\vec{1}^T \Sigma^{-1} \vec{\mu})^2} \right] \vec{\mu} + \left[\frac{[\vec{\mu}^T \Sigma^{-1} \vec{\mu} - r \vec{1}^T \Sigma^{-1} \vec{\mu}]}{(\vec{1}^T \Sigma^{-1} \vec{1})(\vec{\mu}^T \Sigma^{-1} \vec{\mu}) - (\vec{1}^T \Sigma^{-1} \vec{\mu})^2} \right] \vec{1} \right]}$$

This is the portfolio weights for any chosen target return r.

$$\boxed{\sigma_\pi^2(r) = \frac{(\vec{1}^T\Sigma^{-1}\vec{1})r^2 - 2(\vec{1}^T\Sigma^{-1}\vec{\mu})r + \vec{\mu}^T\Sigma^{-1}\vec{\mu}}{(\vec{1}^T\Sigma^{-1}\vec{1})(\vec{\mu}^T\Sigma^{-1}\vec{\mu}) - (\vec{1}^T\Sigma^{-1}\vec{\mu})^2}}$$

so it is in the form $\sigma_\pi^2(r) = ar^2 + br + c$

Plotting return against risk of an n-stock portfolio will give a curve which looks like



How does putting some money in the bank affect the portfolio performance?

Constraint: risky weights no longer sum to 1, bank weight makes up this up

fraction invested in bank

$$\vec{1}\vec{\Lambda} + b = 1$$

Expected return formula:

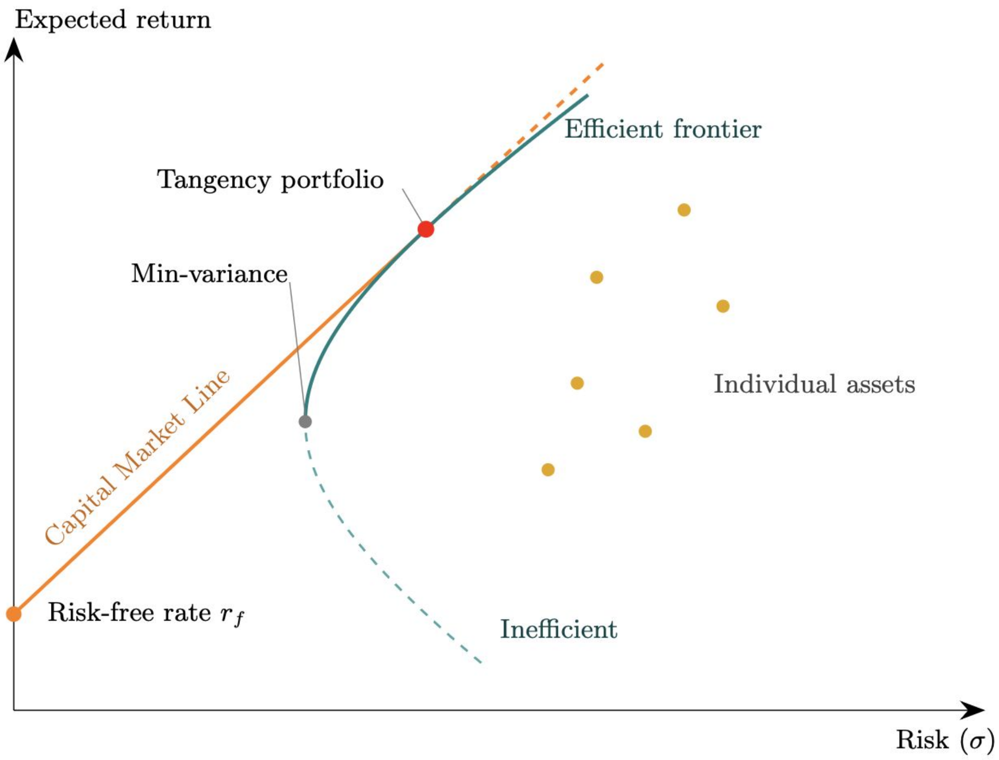
risk free rate of return

$$r = \vec{\mu}^T\vec{\Lambda} + br_f$$

Risk-free asset contributes no variance so risk remains same:

$$\sigma_\pi^2 = \vec{\Lambda}^T\Sigma\vec{\Lambda}$$

Efficient Frontier



Turning point is Global Minimum Variance Portfolio

Slope of CML is

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_\Pi}$$

Aiming to minimise σ_π^2, we can follow the process of Lagrangian optimisation to obtain:

$$L(\lambda_1,\lambda_2,\lambda_3,\alpha,\beta) = \begin{pmatrix}\lambda_1 & \lambda_2 & \lambda_3\end{pmatrix}\begin{pmatrix}1 & 1 & 0\\ 1 & 2 & 1\\ 0 & 1 & 3\end{pmatrix}\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix} + \alpha(\lambda_1+\lambda_2+\lambda_3-1)+\beta(\lambda_1+2\lambda_2+3\lambda_3-r)$$

$$\mathrm{L} = \lambda_1^2+2\lambda_1\lambda_2+2\lambda_2^2+2\lambda_2\lambda_3+3\lambda_3^2+\alpha(\lambda_1+\lambda_2+\lambda_3-1)+\beta(\lambda_1+2\lambda_2+3\lambda_3-r)$$

But finding the gradient and solving it for $\lambda_1, \lambda_2, \lambda_3, \alpha$ & β is long and annoying so…

Example: A stock market with three available stocks

Suppose we know the covariances and the returns for stocks. We want to find the ideal weights for each stock that minimises risk and optimises the sharpe ratio. We have

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

And our constraints are:

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$\lambda_1 + 2\lambda_2 + 3\lambda_3 = r$$

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

...I wrote a python program that takes any 3x3 covariance matrix and expected returns vector and gives the weights for Λ in terms of r and the quadratic in r formed by substituting these values into the risk function for the portfolio

```
l = input("enter a comma separated list of 9 elements in the covariance
returns = input("enter a list of returns for the 3 assets:")
l = list(map(int, l.split(',')))
returns = list(map(int, returns.split(',')))
a = (returns[0] - returns[2]) / (returns[1] - returns[0])
r = sp.var('r')
portfolio = [0]*3
portfolio[1] = -(((2*(r-returns[2]))/(returns[0]-returns[2]))*(l[2]-l[0
portfolio[0] = (r-returns[2]-(returns[1]-returns[2])*portfolio[1])/(ret
portfolio[2] = 1-portfolio[0]-portfolio[1]
y=l[0]*portfolio[0]**2+l[4]*portfolio[1]**2+l[8]*portfolio[2]**2+2*l[1]
print(sp.simplify(y))
print("lambda 1 is", str(portfolio[0]), "lambda 2 is", str(portfolio[1]
```

$$\sigma_\pi^2 = \begin{pmatrix}\lambda_1 & \lambda_2 & \lambda_3\end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix} \begin{pmatrix}\lambda_1 \\ \lambda_2 \\ \lambda_3\end{pmatrix}$$

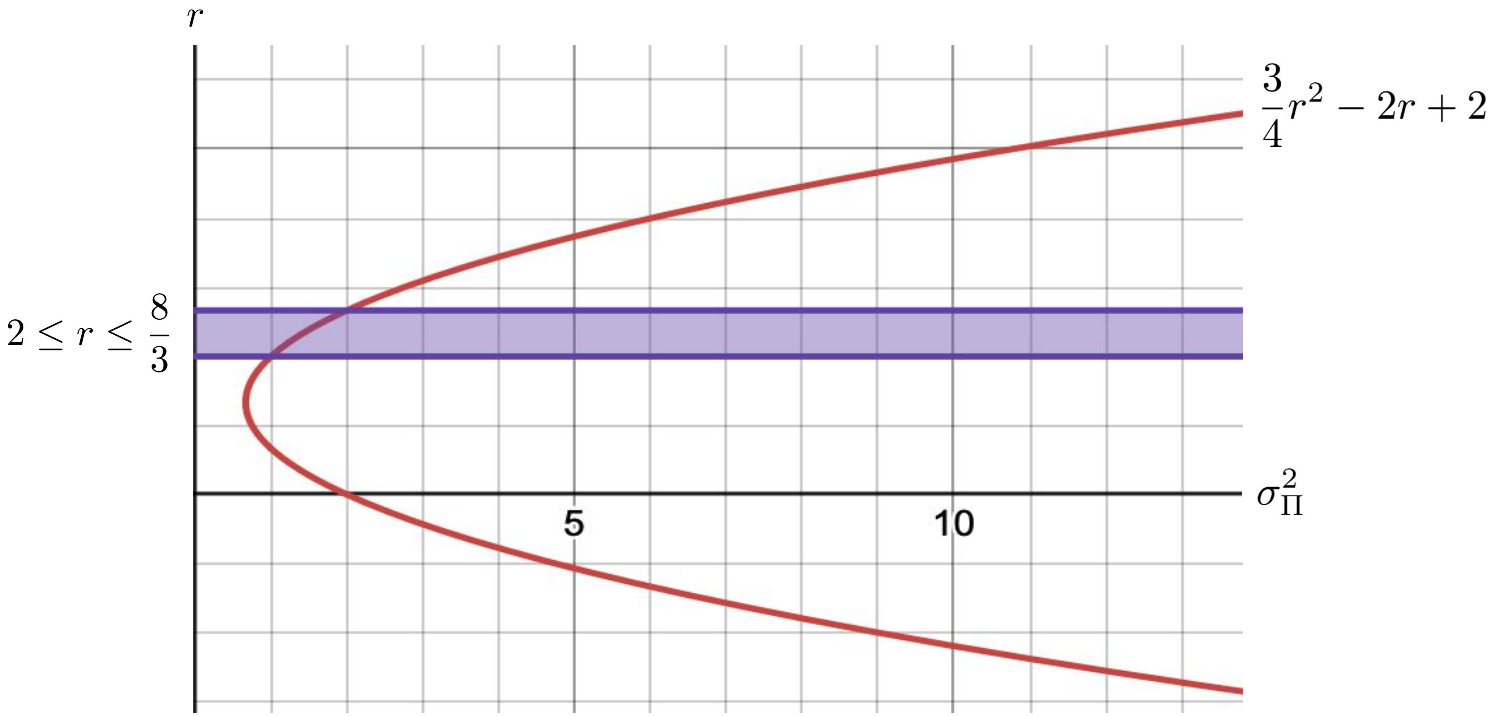
$$\mathbf{\Lambda} = \begin{pmatrix}\lambda_1 \\ \lambda_2 \\ \lambda_3\end{pmatrix} = \begin{pmatrix} 2 - \frac{3}{4}r \\ \frac{1}{2}r - 1 \\ \frac{1}{4}r \end{pmatrix}$$

$$\sigma^2_{\Pi\, min}(r) = \frac{3}{4}r^2 - 2r + 2$$

And imposing non-negative stock allocations results in the inequality

$$\lambda_1, \lambda_2, \lambda_3 \geq 0 \;\Rightarrow 2 \leq r \leq \frac{8}{3}$$

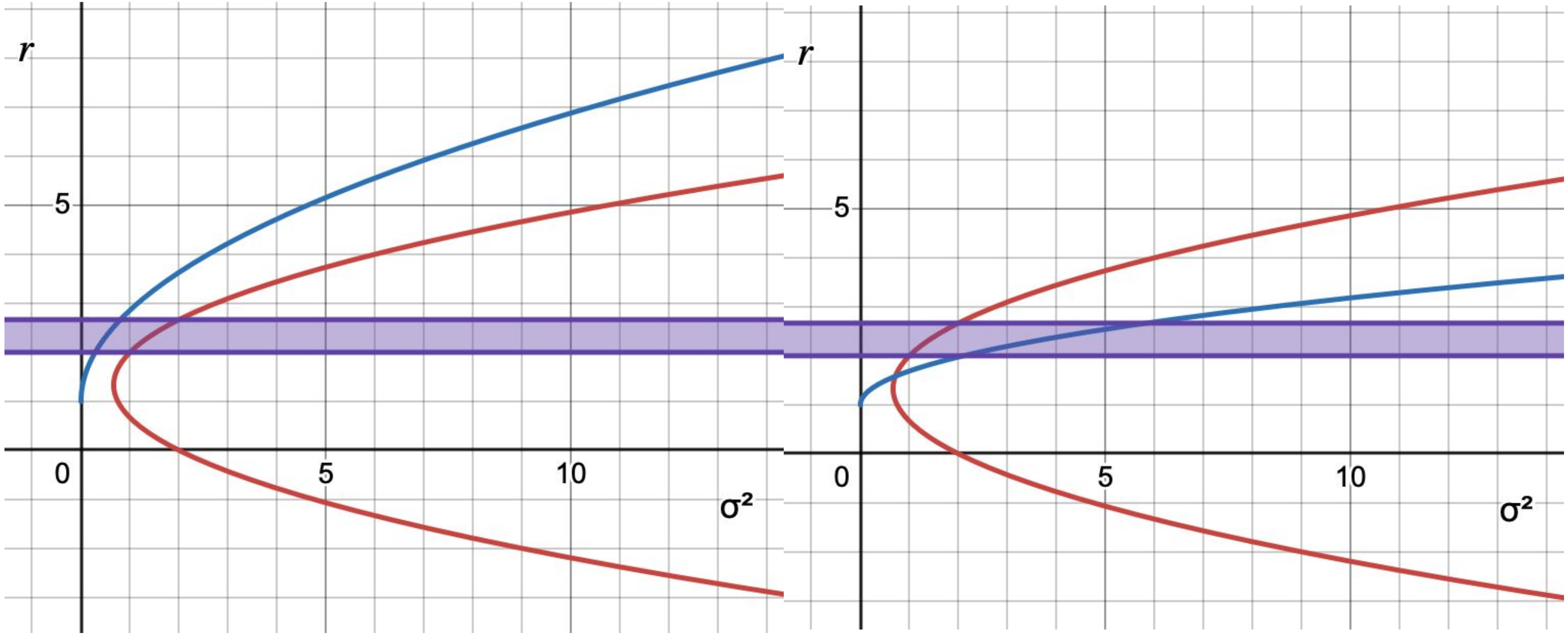
These can be plotted on a graph with risk on the x-axis and return on the y-axis to give us the efficient frontier and the feasible bounds



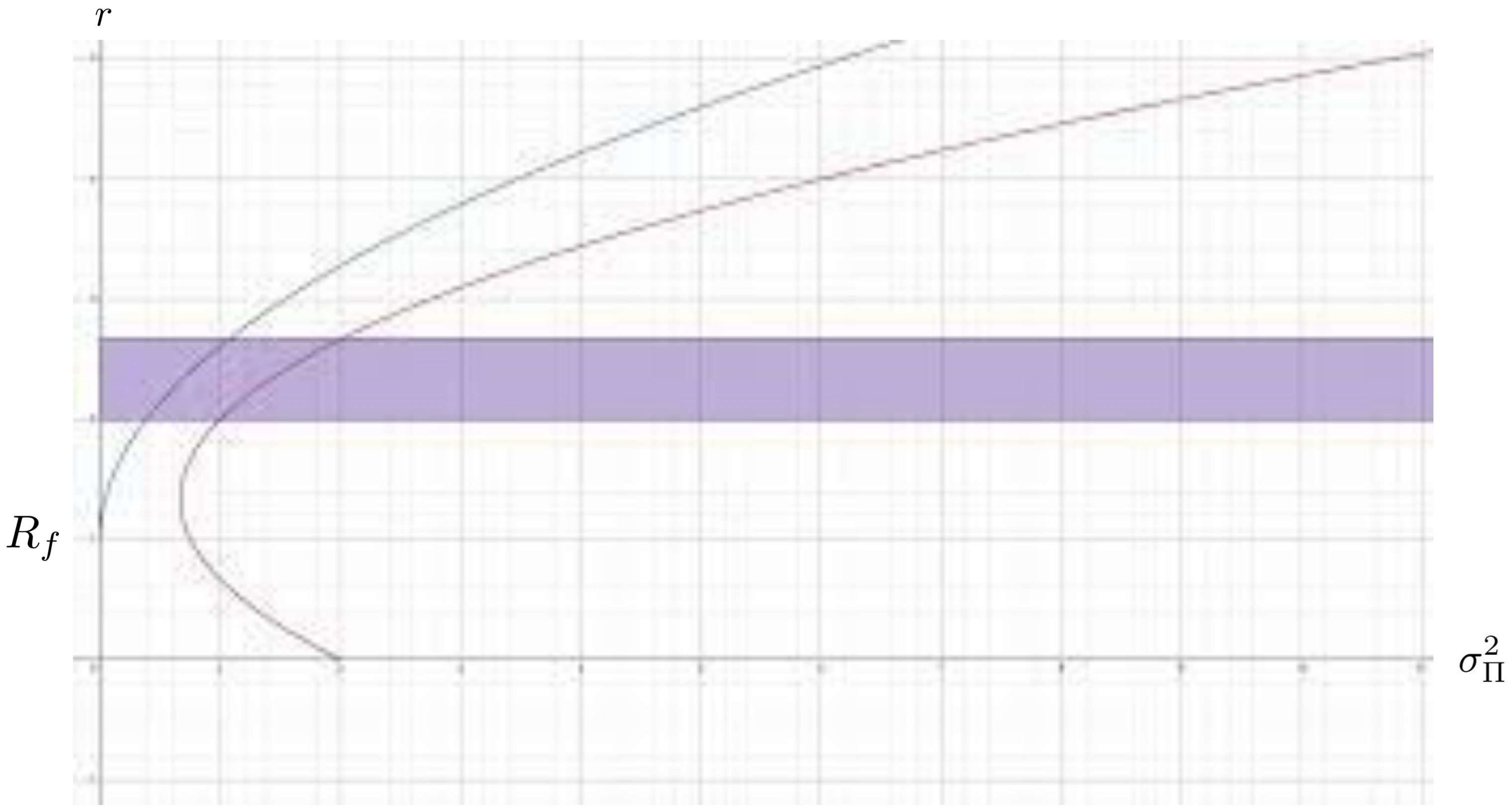
We can also rearrange the Sharpe ratio to form a curve which we can plot for different values of s

$$s := Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_\Pi} \Rightarrow R_p = s\sigma_\Pi + R_f$$

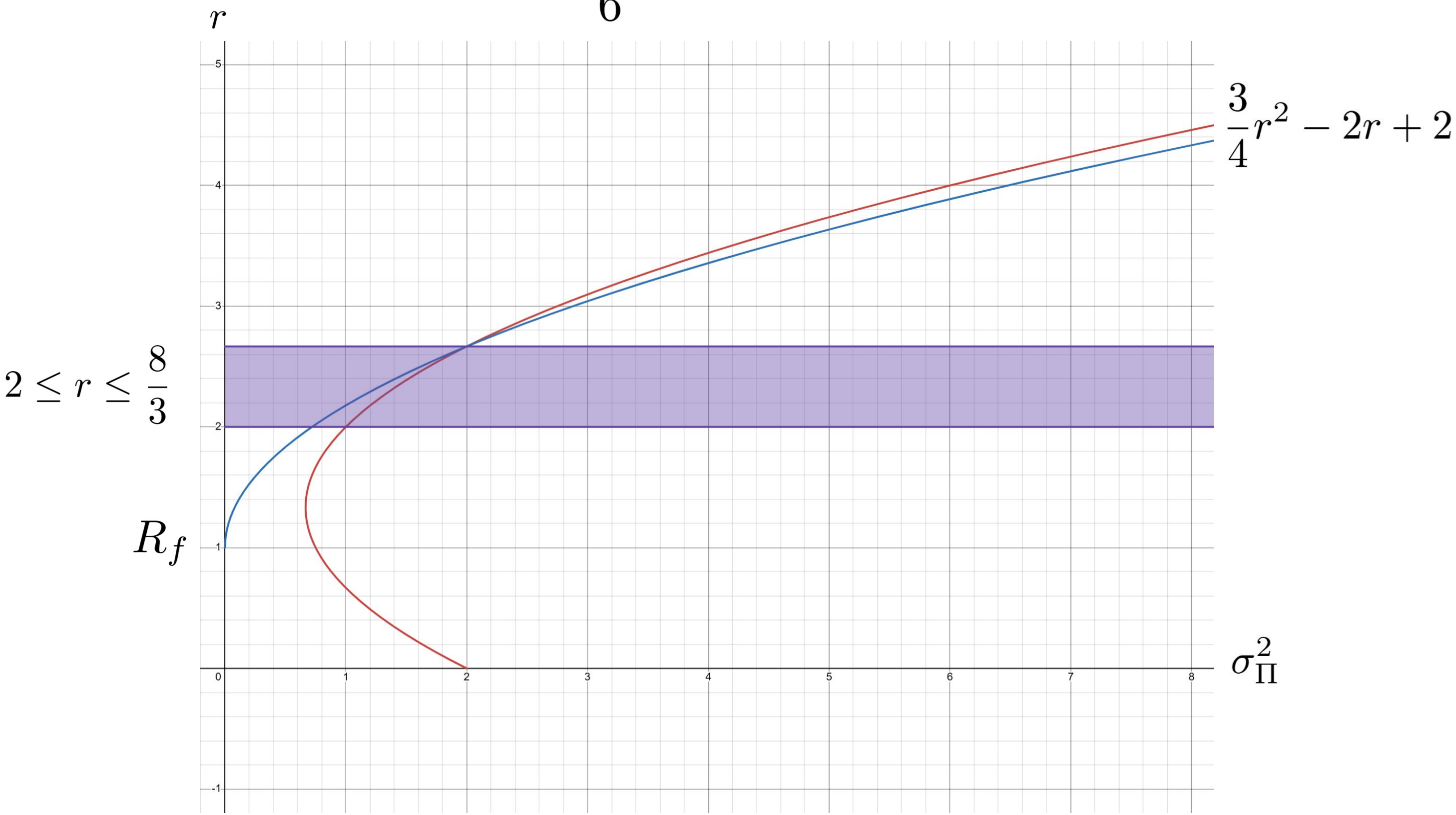
https://www.desmos.com/calculator/szk1zi9akm



Let $s := Sharpe\ Ratio = \dfrac{R_p - R_f}{\sigma_\Pi} \quad \Rightarrow \quad R_p = s\sigma_\Pi + R_f \ (= s\sqrt{\sigma_\Pi^2} + R_f)$



If we set $R_f = 1$ we find the optimal Sharpe ratio occurs where r is maximised at $r = \frac{8}{3}$ which gives us Sharpe ratio $s = \frac{5\sqrt{2}}{6} \approx 1.18$ and the allocation $\lambda_1 = 0,\ \lambda_2 = \frac{1}{3},\ \lambda_3 = \frac{2}{3}$



From Theory to Practice

How Portfolio Theory is Used at Goldman Sachs XIG

Source: Goldman Sachs XIG Recruitment Webinar, June 2026

How XIG Uses Portfolio Theory in Practice

During the Goldman Sachs XIG Recruitment Webinar, we asked how concepts such as utility functions, covariance matrices, Sharpe ratios and portfolio optimisation are applied in real investment decisions.

The speakers explained that these tools are widely used as the quantitative foundation of portfolio construction.

Academic Concept	Practical Use in XIG
Sharpe Ratio	Screen hedge funds and evaluate risk-adjusted returns
Covariance Matrix	Measure relationships between funds and improve diversification
Efficient Frontier	Identify attractive risk-return combinations
Utility Function	Match portfolios to different client risk preferences

Quantitative Models Are Only the Starting Point

XIG emphasised that portfolio optimisation models are powerful tools, but they are not sufficient on their own.

After quantitative screening, investment professionals still evaluate:

• Fund manager quality

•Liquidity constraints

• Different market environments

• Client preferences

• Quantitative models may not fully capture future market shocks

Therefore, successful investing requires both quantitative analysis and qualitative judgement.

Final Conclusion

The goal of portfolio optimisation is not to eliminate risk, but to manage it through diversification and align investments with an investor’s objectives and risk preferences.

In other words:

Don’t put all your eggs in one basket.